

Representing Cancer Mutations using Knowledge Graphs

Ulli (Ulrike) Steindl*, i6369824

March 31, 2024

Abstract

Mutational signatures are a data structure in cancer research in the domain of genomics which has allowed researchers to describe and subclassify mutations in the human body. This finer classification has brought about new empirical findings and subgroups. The current project authored an ontology for a part of this space, and has constructed a knowledge graph representation for it. This allows for conclusions that are otherwise hard draw in this complex entity space. This work is intended as a proof of concept.

1 Introduction

Mutational Processes in the human genome have been studied from a variety of different perspectives: which mutagenes cause which types of mutation at which location in the genome, which types of cancers do they occur in, which treatment is most successful? With the increased availability of genomic data, new methods of quantifying mutational processes have developed that are based on comparing the DNA of mutated and non-mutated tissue on a large scale. Information about the contexts, strengths and types of these mutations have been divided into subtypes called Mutational Signatures (MS) using non-negative matrix factorization [Alexandrov et al., 2020]. The COSMIC database in its current version v3.4 curated several hundred different subtypes of mutation patterns and maintains links to particular etiologies, as well as providing relevant literature [Sondka et al., 2023].

MS have proven instrumental in subclassifying cancer types, and improving the recommendations for more specific treatments that show higher effectivity. Specific classes of DNA mismatch repair were linked with certain mutational signatures [Van Hoeck et al., 2019], which in turn respond better to a particular treatment class.

Research in cancer and mutation types identifying previously unknown generalizations are an important stepstone to understanding cancer in its manifold representations. The relations between different environmental factors and entities are notoriously complex, and difficult to wrangle. Researchers in the field need to have very good knowledge of literature in order to be able to make complex inferences and queries. Constructing a knowledge graph of these complex phenomena will help make sophisticated inferences, and provide a structure for academic research in a thriving area of genomics.

The current project authored an ontology covering the entity domain of mutational signatures, and has constructed a knowledge graph of the SBS signature type of the COSMIC dataset in version v3.4. It took into account a number known connections between mutational signatures, mutations, etiologies, tumor types, body parts, and sample information, in addition to including some references.

*I am a research assistant at the research group for [Computational Cancer Biology](#), a joint group at the University Clinics Aachen and Cologne, and this work is a part of this research agenda, and may continue to be developed after the completion of this project. I will highlight some of the possible future avenues myself in the project report, but I would be curious to hear from you, gentle reader, about where you think a promising direction could be.

2 Related Work

A standardized version of Mutational Signatures coming from an accepted pipeline are released under creative commons license by [Sondka et al. \[2023\]](#). The semi-structured data based on the Cosmic database v.3.4 [[Sondka et al., 2023](#)] will be mined and form the core the current knowledge graph. Other signature models and types are also discussed in the community, as [Degasperi et al. \[2020\]](#), but are less suited to drawing quantificational conclusions. Consequently, my attention for this project will focus on the signatures presented in the COSMIC format.

Ontologies in the general space of oncology are already available. For the current project, I strive for semantic interoperability with the Ontologies and Vocabularies defined in the Thesaurus of the National Cancer Institute [[NCI, 2024](#)]. I am unaware of any ontologies covering the space of MS, so developing this part of the ontology in the current project will be a novel contribution.

To represent the localities in genomic information, the Sequence ontology was created [Eilbeck et al. \[2005\]](#). When it is necessary to refer to particular locations in the genome, I will refer to this ontology.

Functional genetic information is captured in the Gene Ontology [Aleksander et al. \[2023\]](#), [Ashburner et al. \[2000\]](#). I will strive for interoperability with this ontology when working in this domain.

3 Methodology

The proposed project constructed an ontology and a knowledge graph representation in RDF using the OWL specification, modeling relations between mutations, mutational signatures, cancer types, etiologies, treatments, and literature sources. This was done using the *protegé* software [Musen \[2015\]](#), which provided an important tool in structuring and visualizing the task. The base of this graph is <https://computationalcancerbiology.uni-koeln.de/mso#>, which is not currently an active domain, but may become so in the future.

The project compiled data from these two structured and semi-structured sources: the review article [Alexandrov et al. \[2020\]](#), and the COSMIC database itself [Sondka et al. \[2023\]](#). In the case of the former, the data was extracted from the supporting documents, in the case of the latter, relevant information was scraped off their website.

The resulting knowledge graph will be tested by deploying a local end point, and querying it with SPARQL queries that are relevant for academic discovery. The queries are presented in the following section.

4 Results

4.1 Ontology Design

In total, 133 owl classes and subclasses were defined, as well as 91 properties. In the following sections, I will describe some of the more important items in the semantic space of mutational signatures.

4.1.1 The Mutational Signature Classes

Visualization of the relations between Mutational Signatures and its neighbors is given in Figure 1 in the appendix. `Mutational Signature` is a sub-class of `Observation`, which houses most of the numerical values in the `mso` graph. The mutational signature class has six subClasses,

as there are different mutation types, and for each of them, different signatures are calculated. Since most of the research in the area focuses on the mutation type of a single base substitution (SBS), the current work has started by fully implementing this part of the ontology. A single base substitution is represented as a substitution of the middle base in a 3-mer context, for example, in the substitution A[C>A]A, which is represented using the convention A.C-A.A in the mso knowledge graph, the base C is replaced by A in a context where A precedes and A follows the mutation site.

To calculate the mutational signatures, a mutational catalogue V of counts of such mutations is compiled from a number of reference genomes. The class for the data in the mutational catalogue is `MutationCatalogue`. This catalogue is then factorized into the matrices W and H , where H is represented in the class `MutationalExposure`, and W is finally captured as the `MutationalSignature` class. The features of samples are captured in the `Sample` class, with subclass `ReferenceSample` for the named genomes that are used in the calculation of the signatures that are published on the COSMIC homepage. It includes their name and provenience. Using the Object Property `COSMIC_uses_sample`, the sample instances and Mutational Signatures are linked.

The `MutationalSignature` class is divided into subtypes SBS, (single base substitution), DBS (doublet base substitution, for substitutions of two adjacent bases), ID (small insertions and deletions), CN (copy number variations), and SV signatures. The current draft of the mso graph focused on implementing the SBS subtype.

Single base mutations can happen in 96 contexts, an example graph is shown in Figure 2. Each of row labels was implemented as a class with its own name and label (e.g. A.C-A.A), as a subclass of Class `SBS_probability`, which is in turn a subclass of Class `MutationalProbability`. These instances are linked to the `MutationalSignature` classes using the Object Properties `has_part` and `is_part` which was implemented following the recommendation in Rector et al. [2005].

Finally, the Data property `hasProbabilityValue` links the Mutational Probabilities to its Literal values, which are implemented with the appropriate restrictions.

The code to add this data to the rdf graph is contained in the section `Signatures and Samples` of the attached jupyter notebook `Constructing_MSO.ipynb`. The signature data was processed using the `get_signatures_from_soup`

4.1.2 Cosmic properties

On the COSMIC homepage, each of the signatures is further described qualitatively in a table. Such a table is pictured in Figure 3 in the appendix. The properties contained in this section were represented in a set of predicates each prefixed with `COSMIC_`. This included the predicates listed below, which had either string or boolean domains, the class `COSMIC_version`, or doi objects in the case of literature as their domain. As the table also involved “evidence strength” as a factor, they were realized as instances of these predicates which also linked to the `COSMIC_evidence` class.

COSMIC_properties Class for all COSMIC properties

COSMIC_first_included_in When a signature was first included in the COSMIC set, e.g. ‘v1’. There is ongoing discussion about how many signatures there should be.

COSMIC_has_version Which version does a signature belong to? E.g. ‘v3.4’

COSMIC_identification_study Study where this signature was introduced. Domain: a doi object

COSMIC_uses_sample E.g. a reference genome like GrCh37, implemented using the `Sample` class

COSMIC_obj_properties_with_evidence_strength Superclass for properties that are parametrized using evidence strength.

COSMIC_aetiology What may have caused the mutation? A string literal. At a later point in time, this should be replaced by an entity in a related knowledge graph like NCIT.

COSMIC_experimental_evidence Studies that were conducted. A doi object.

COSMIC_experimental_validation_species Links to instances of the *Species* class

COSMIC_experimentally_validated Was this signature experimentally confirmed?
Links to a boolean literal

COSMIC technically_validated Was the signature technically validated? Links to a boolean literal

The data was collected using the code in the functions `url_to_soup`, `get_table_from_soup` in the attached jupyter notebook `Constructing_MSO.ipynb`. The code in section `Cosmic Properties` of the notebook maps the data to rdf.

4.1.3 Mutation and Location

Classes for *Mutation* and *Location* were implemented. While currently not filled with instances, a future plan is to link Mutational signatures to concrete instances of mutations that happen at a particular location in a genome. This location may be a cytogenetic location as a particular point, a range, or also an exon, which in this ontology at current is conceptualized as a named DNA range. Although whole genome sequencing data is becoming increasingly available, it is worthwhile to also think of the concept of mutational sequences for subsamples of DNA, and more samples are available of regions containing exons than introns. These relations are modeled using the `is_at` object property. In a medium-term perspective, I would like to fill these classes with instances and model in a more detailed fashion, should the project continue beyond the scope of the classwork. If the development continues, it is desirable to use the Sequence Ontology, or its subset, the Sequence Ontology Feature Annotation (SOFA) [Eilbeck et al. \[2005\]](#) to model genetic sequencing data.

4.1.4 Tumors and Body Parts

An interesting data set is presented in [Figure 3](#) in the study [Alexandrov et al. \[2020\]](#). It presents the proportion of tumors containing a given signatures, as well as the median mutational burden covered by each of the signatures. It is just one example the research agenda to link cancers with mutational signatures rather than types of sicknesses.

This data is represented in the mso graph using the predicates `has_signature` with subclass `has_signature_proportion`, which has subclasses `has_signature_proportion_{number}`. I opted for this fine-grained implementation to allow for the user to filter for the proportion value. These predicates link tumors implemented in Class *Tumor*, which contains instances of the NCIT ontology [\[NCI, 2024\]](#). These in turn were linked to the body parts and organs they occur in using the *BodyPart* class.

In a similar manner, the properties `has_mutations`, `has_mutations_in` and `has_{number}_mutations_in` were implemented to represent the mutational burden.

This code can be found in the `Tumor association` section of the enclosed ipython notebook.

4.1.5 Treatment and Aetiology

There is increasing research about treatments being particularly effective or ineffective for cancer types of a particular mutation, summarized e.g. by [Van Hoeck et al., 2019](#). Due to this fact,

it is also interesting to study the connection between Treatments, cancer types, and mutational signatures. The current knowledge graph also does not include data regarding treatment, although in the future these items should be linked to entities in external entities, like the ones in [NCI \[2024\]](#).

4.2 Testing the Knowledge Graph

4.2.1 For a given mutational signature, which cancer is it associated with?

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX mso: <https://computationalcancerbiology.uni-koeln.de/mso#>
PREFIX obo: <http://purl.obolibrary.org/obo/>

SELECT DISTINCT ?obj ?cancertype WHERE {
  ?sub ?pred ?obj .
  ?sub rdfs:label ?cancertype .
  FILTER (REGEX(STR(?pred), "has_signature"))
}
ORDER BY ?obj ?cancertype
LIMIT 30
```

The result is presented in Table 1 in the appendix

4.2.2 List the mutational signatures that are experimentally validated?

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX mso: <https://computationalcancerbiology.uni-koeln.de/mso#>
PREFIX obo: <http://purl.obolibrary.org/obo/>

SELECT DISTINCT ?sub WHERE {
  ?sub ?pred ?obj .
  FILTER (REGEX(STR(?pred), "COSMIC_experimental_evidence"))
}
ORDER BY ?sub
```

The result is presented in Table 2 in the appendix

4.2.3 Of the signatures that are associated with Bladder cancer, which five SBS probabilities are the highest?

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX mso: <https://computationalcancerbiology.uni-koeln.de/mso#>
PREFIX obo: <http://purl.obolibrary.org/obo/>
SELECT DISTINCT ?value ?prob_label ?sig WHERE {
  {
    ?sig mso:has_part ?prob .
    ?prob rdfs:label ?prob_label .
    ?prob mso:hasProbabilityValue ?value .
  }
```

```

    ?cancer ?pred ?sig .
    ?cancer rdfs:label "Bladder-TCC"@en .
    FILTER (REGEX(STR(?pred), "has_signature"))
  }
}
ORDER BY DESC(?value)
LIMIT 5

```

The results are presented in Table 3 in the appendix.

5 Discussion

In my work on the mso graph, the aim was to model in as detailed a fashion as possible. The modeling was accurate and represented the data in the expected way. This work cannot exceed the scope of a proof of concept, however.

As it is a proof of concept, many of the action points for possible future development become immediately clear.

It is necessary to collect, process and aggregate more data to fill in the regions of the knowledge graph that are currently only sketched: fully integrating genomic location data into the ontology, finding and processing data with structured treatment information and connecting it to mutational signatures, modeling all types of mutational signatures, maybe in different versions, and testing exhaustively. Clearly much work can still be done in compiling the graph.

After complete generation, an exciting avenue may be knowledge generation of in the graph. Could connections be identified that are not evident in the sources of the graph? Could statistical learning be applied using graph embeddings that may produce results that are not attainable via conventional machine learning models based on the matrix representations of mutational signatures?

6 Conclusions

The current work provided a proof of concept for constructing a knowledge graph in the domain of mutational signatures. It authored an ontology for the domain, and collected structured and semi-structured data from homepages and academic publications, which were then used to populate the graph. It was found that the resulting graph could draw advanced conclusions from these two data sources, and can be helpful in knowledge generation.

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- nikov, C. Smith, A. Cuzick, J. Seager, L. Cooper, J. Elser, P. Jaiswal, P. Gupta, P. Jaiswal, S. Naithani, M. Lera-Ramirez, K. Rutherford, V. Wood, J. L. De Pons, M. R. Dwinell, G. T. Hayman, M. L. Kaldunski, A. E. Kwitek, S. J. F. Laulederkind, M. A. Tutaj, M. Vedi, S.-J. Wang, P. D'Eustachio, L. Aimo, K. Axelsen, A. Bridge, N. Hyka-Nouspikel, A. Morgat, S. A. Aleksander, J. M. Cherry, S. R. Engel, K. Karra, S. R. Miyasato, R. S. Nash, M. S. Skrzypek, S. Weng, E. D. Wong, E. Bakker, T. Z. Berardini, L. Reiser, A. Auchincloss, K. Axelsen, G. Argoud-Puy, M.-C. Blatter, E. Boutet, L. Breuza, A. Bridge, C. Casals-Casas, E. Coudert, A. Estreicher, M. Livia Famiglietti, M. Feuermann, A. Gos, N. Gruaz-Gumowski, C. Hulo, N. Hyka-Nouspikel, F. Jungo, P. Le Mercier, D. Lieberherr, P. Masson, A. Morgat, I. Pedruzzi, L. Pourcel, S. Poux, C. Rivoire, S. Sundaram, A. Bateman, E. Bowler-Barnett, H. Bye-A-Jee, P. Denny, A. Ignatchenko, R. Ishtiaq, A. Lock, Y. Lussi, M. Magrane, M. J. Martin, S. Orchard, P. Raposo, E. Speretta, N. Tyagi, K. Warner, R. Zaru, A. D. Diehl, R. Lee, J. Chan, S. Diamantakis, D. Raciti, M. Zarowiecki, M. Fisher, C. James-Zorn, V. Ponferrada, A. Zorn, S. Ramachandran, L. Ruzicka, M. Westerfield, S. A. Aleksander, J. Balhoff, S. Carbon, J. M. Cherry, H. J. Drabkin, D. Ebert, M. Feuermann, P. Gaudet, N. L. Harris, D. P. Hill, R. Lee, H. Mi, S. Moxon, C. J. Mungall, A. Muruganugan, T. Mushayahama, P. W. Sternberg, P. D. Thomas, K. Van Auken, J. Ramsey, D. A. Siegele, R. L. Chisholm, P. Fey, M. C. Aspromonte, M. V. Nugnes, F. Quaglia, S. Tosatto, M. Giglio, S. Nadendla, G. Antonazzo, H. Attrill, G. dos Santos, S. Marygold, V. Strelets, C. J. Tabone, J. Thurmond, P. Zhou, S. H. Ahmed, P. Asanithong, D. Luna Buitrago, M. N. Erdol, M. C. Gage, M. Ali Kadhum, K. Y. C. Li, M. Long, A. Michalak, A. Pesala, A. Pritazahra, S. C. C. Saverimuttu, R. Su, K. E. Thurlow, R. C. Lovering, C. Logie, S. Oliferenko, J. Blake, K. Christie, L. Corbani, M. E. Dolan, H. J. Drabkin, D. P. Hill, L. Ni, D. Sitnikov, C. Smith, A. Cuzick, J. Seager, L. Cooper, J. Elser, P. Jaiswal, P. Gupta, P. Jaiswal, S. Naithani, M. Lera-Ramirez, K. Rutherford, V. Wood, J. L. De Pons, M. R. Dwinell, G. T. Hayman, M. L. Kaldunski, A. E. Kwitek, S. J. F. Laulederkind, M. A. Tutaj, M. Vedi, S.-J. Wang, P. D'Eustachio, L. Aimo, K. Axelsen, A. Bridge, N. Hyka-Nouspikel, A. Morgat, S. A. Aleksander, J. M. Cherry, S. R. Engel, K. Karra, S. R. Miyasato, R. S. Nash, M. S. Skrzypek, S. Weng, E. D. Wong, E. Bakker, T. Z. Berardini, L. Reiser, A. Auchincloss, K. Axelsen, G. Argoud-Puy, M.-C. Blatter, E. Boutet, L. Breuza, A. Bridge, C. Casals-Casas, E. Coudert, A. Estreicher, M. Livia Famiglietti, M. Feuermann, A. Gos, N. Gruaz-Gumowski, C. Hulo, N. Hyka-Nouspikel, F. Jungo, P. Le Mercier, D. Lieberherr, P. Masson, A. Morgat, I. Pedruzzi, L. Pourcel, S. Poux, C. Rivoire, S. Sundaram, A. Bateman, E. Bowler-Barnett, H. Bye-A-Jee, P. Denny, A. Ignatchenko, R. Ishtiaq, A. Lock, Y. Lussi, M. Magrane, M. J. Martin, S. Orchard, P. Raposo, E. Speretta, N. Tyagi, K. Warner, R. Zaru, A. D. Diehl, R. Lee, J. Chan, S. Diamantakis, D. Raciti, M. Zarowiecki, M. Fisher, C. James-Zorn, V. Ponferrada, A. Zorn, S. Ramachandran, L. Ruzicka, and M. Westerfield. The gene ontology knowledgebase in 2023. *GENETICS*, 224(1), Mar. 2023. ISSN 1943-2631. doi: 10.1093/genetics/iyad031.
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7 Appendix

Table 1: Results for query in section 4.2.1

obj	cancertype
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Biliary-AdenoCA
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Bladder-TCC
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Bone-Osteosarc
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Bone-Other
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Breast
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	CNS-GBM
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	CNS-Medullo
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	CNS-Oligo
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	CNS-PiloAstro
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Cervix
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Colorect-AdenoCA
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Eso-AdenoCA

https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Head-SCC
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Kidney-ChRCC
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Kidney-RCC
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Liver-HCC
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Lung-AdenoCA
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Lung-SCC
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Lymph-BNHL
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Lymph-CLL
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Myeloid-AML
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Myeloid-MDS/MPN
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Ovary-AdenoCA
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Panc-AdenoCA
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Panc-Endocrine
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Prost-AdenoCA
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Skin-Melanoma
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Soft tissue-Leiomyo
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Soft tissue-Liposarc
https://computationalcancerbiology.uni-koeln.de/mso#sbs1	Stomach-AdenoCA

Table 2: Query results from section 4.2.2

sub
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs10a
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs10b
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs11
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs13
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs14
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs15
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs17b
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs18
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs2
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs20
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs21
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs22a
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs24
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs26
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs28
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs3
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs30
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs31
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs35
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs36
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs4
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs42
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs44
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs45
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs6
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs7a
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs7b
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs87

https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs88
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs90
https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs99

Table 3: Results from section [4.2.3](#)

value	prob_label	sig
0.536015480663095	T.C-T.A	https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs2
0.535130188298692	T.C-T.A	https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs2
0.512074127439389	T.C-T.A	https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs2
0.51207199621427	T.C-T.A	https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs2
0.511533628982139	T.C-T.A	https://computationalcancerbiology.uni-koeln.de/mso#SBS_v3.4_sbs2

Artificial Intelligence Usage

ChatGPT 3.5 was used as a coding assistant for various parts of the project. It has been my experience that what it provides you usually needs further development and debugging, and serves more as a memory aid than a tool to do things I cannot do.

Acceptance criteria

Supporting evidence for mutational signature validity

Validated evidence for *real signature*

Unclear evidence for *real signature*

Evidence for *artefact signature*

Background	Identification study		First included in COSMIC
	Alexandrov <i>et al.</i> , 2013 <i>Nature</i>		v1
Identification	NGS technique	Different variant callers	Multiple sequencing centres
	WES & WGS	Yes	Yes
Technical validation	Validated in orthogonal techniques	Replicated in additional studies	Extended context enrichment
	Yes	Yes	-
Proposed aetiology	Mutational process		Support
	Tobacco smoking		Experimental confirmation
Experimental validation	Experimental study		Species
	Nik-Zainal <i>et al.</i>, 2015 <i>Mutagenesis</i>		Mouse

Summary of the technical and experimental evidence available in the scientific literature regarding the validation of the mutational signature.

Figure 3: Example of an "Acceptance Criteria" section for sbs4